Name : .....

Student ID: .....

Signature: .....

The University of New South Wales

School of Electrical Engineering and Telecommunications

Final Examination

Session 1 2013

ELEC3104: Digital Signal Processing

Candidates should attempt ALL questions

1. Reading time: 10 minutes
2. Time allowed: 3 hours.
3. The paper contains 5 questions – answer all questions.
4. The questions are worth 100 marks in total.
5. Print your name, student ID and question number on the front page of each answer book.
6. This exam contributes 60% of the total course assessment.
7. The questions are NOT of equal value. Part marks are as indicated.
8. The examination paper may not be retained by the candidate.
9. This is a closed book exam.
10. The use of UNSW approved calculators is permitted.
11. Transform tables are attached to this exam paper for your use.
12. This paper contains 7 pages apart from this instruction page.

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK.

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Information sheet

Table of the z transforms of useful sequences

Basic z-Transforms	Sequence	z-Transform
1. Unit sample	$\delta[n]$	1
	$\delta[n-k]$	z^{-k}
2. Unit step	$u[n]$	$z/(z-1)$
3. Exponential	$a^n u[(n)]$	$z/(z-a)$
4. Sinusoidal	$\sin[\theta_o n]\, u[n]$	$\dfrac{z \sin\theta_0}{z^2 - 2\cos\theta_0\, z + 1}$
	$\cos[\theta_o n]\, u[n]$	$\dfrac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0\, z + 1}$
5. Unit ramp	$nu[n]$	$\dfrac{z}{(z-1)^2}$

$$1 + z^{-1} + z^{-2} + z^{-3} + \ldots\ldots\ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

z-Transform Properties	Sequence	z-Transform
	$x[n]$	$X(z)$
	$x[n-k]$	$z^{-k} X(z)$
	$a^n x[n]$	$X(z/a)$
	$x[-n]$	$X(z^{-1})$
	$x[n]*y[n]$	$X(z).Y(z)$
	$n\, x[n]$	$-z\,(dX(z)/dz)$

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\cos\theta = \frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right]$$
$$\sin\theta = \frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$$

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Information Sheet

Fourier Series:

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t); \qquad \omega_0 = 2\pi f_0; \quad f_0 = \frac{1}{T}$$

The fundamental frequency of x(t) is ω_0 and T is the period of the signal x(t).

$$A_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\omega_0 t)\,dt \qquad n = 0,1,2.... \qquad B_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\omega_0 t)\,dt \qquad n = 1,2,3...$$

Fourier Transform pair (for discrete-time signal)

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}... \qquad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

θ = *digital frequency*; $-\pi \le \theta \le \pi$

Fourier Transform pair (for continuous-time signal)

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega$$

A **Butterworth filter** of order 'n' is defined by

$$|H(j\omega)|^2 = \frac{1}{1+\left(\frac{\omega}{\omega_c}\right)^{2n}}$$ where ω_c is the cut-off frequency of the filter

n	Butterworth Polynomials
1	$S+1$
2	$s^2+\sqrt{2}s+1$
3	$(s^2+s+1)(s+1)$

Impulse Invariant Method:

$$\frac{1}{s+b} \rightarrow \frac{1}{1-e^{-bT}z^{-1}} \qquad b>0 \quad T\text{-sampling period}$$

Bilinear Transformation:

$$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} \qquad T - sampling\ \ period$$

Probability Density Function (PDF):

The quantisation error (e) is normally assumed to be random and uniformly distributed with zero mean. Therefore, the PDF, $P(e) = 1/\Delta V$, where ΔV is the quantisation step size.

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Question 1 (19 marks)

(a) If an analogue signal $x(t)$ is sampled every T seconds, then at the output of the Analogue to Digital Converter, as shown in Fig 1(a), we obtain a discrete-time signal $x[n]$; where t = time; n = sample number; ω = analogue frequency; θ = digital frequency; $X(\theta)$ = digital spectrum; $X(\omega)$ = analogue spectrum, k = an integer

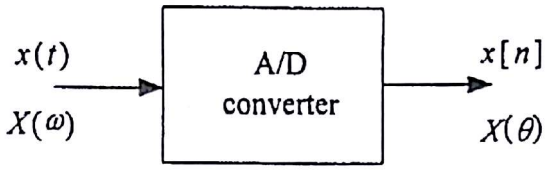


Figure 1(a)

Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta)=\frac{1}{T}\sum_{k=-\infty}^{\infty}X(\omega+k\frac{2\pi}{T}).$$

[5 marks]

(b) With the aid of simple spectral diagrams, explain how aliasing distortion could arise if analogue signals are not low-pass filtered before they are converted to discrete time signals. You may use the equation derived in part (a) above to answer this question. [4 marks]

(c) Consider the analogue signal,

$$x(t) = \sin(900\pi t) + 2\sin(1700\pi t) + \sin(4200\pi t)$$

(i) If we sample the signal $x(t)$ at a sampling rate of 1 kHz, what is the discrete-time signal $x[n]$ obtained after sampling? [2 marks]

(ii) What is the analogue signal $y(t)$ that we can reconstruct from the samples obtained in part (i) above, if we use an ideal interpolation? [3 marks]

(d) Let a causal linear time invariant discrete-time system be characterised by a real impulse response $h[n]$ with a DTFT of $H(\theta)$. Consider the system shown in Figure 1(b), where $x[n]$ is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ in terms of $H(\theta)$, and show that it has a zero-phase response. [5 marks]

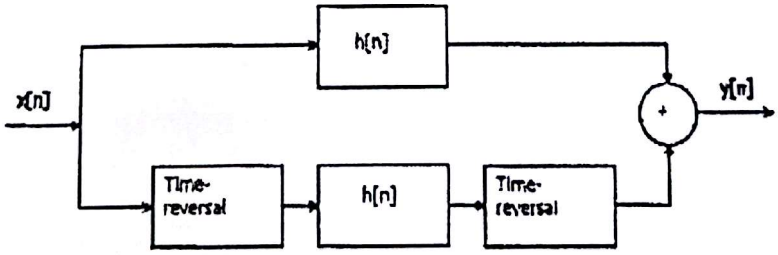


Figure 1(b)

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Question 2 (18 marks)

(a) Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

$$SQNR = 6.02B + 4.77 - 20\log(\frac{A}{\sigma_{sig}})$$

where the input range of the A/D converter is ±A and the rms value of the input signal is σ_{sig}. Determine the SQNR if B is 16 bits and the input is:

(i) a sine wave

(ii) a signal with an rms value of 0.2A

[6 marks]

(b) The following first-order IIR digital filter has been proposed for an application:

$$H(z) = k\frac{1 - z^{-1}}{1 + b\,z^{-1}} \qquad 0 < b < 1$$

(i) Determine the value of k in the transfer function $H(z)$, so that the filter has a 0 db gain at $\theta = \pi$; where θ is the digital frequency.

[2 marks]

(ii) Using the value of k from part (i) above, determine the magnitude response of the transfer function $H(z)$ and show that it has a high pass response.

[3 marks]

(iii) Show that the 3dB cut-off frequency (θ_c) of the transfer function $H(z)$ from part (i) above, is given by

$$\theta_c = \cos^{-1}(\frac{-2b}{1 + b^2})$$

[4 marks]

(c) Show that the following transfer function can be obtained using a parallel connection of two all-pass filters.

$$H(z) = \frac{14 + 11z^{-1}}{9 + 6z^{-1}}$$

[3 marks]

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Question 3 (23 marks)

(a) (i) Determine and sketch the magnitude and phase response of the following filter:

$$y[n] = x[n] - x[n-8].$$ [4 marks]

(ii) Show that the above filter has constant group delay. [2 marks]

(b)

(i) Determine the transfer function $H(z)$ of a discrete-time system with the pole-zero map given in Figure 2. [3 marks]

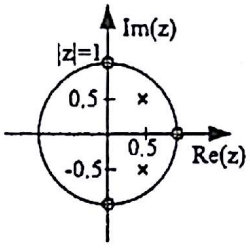


Figure 2

(ii) Sketch an approximate magnitude response from the pole-zero map given in Figure 2. [3 marks]

(c) In applications involving the digital modulation of sinusoidal carrier signals, it is necessary to generate sine and cosine signals simultaneously from an oscillator. Consider the system

$$y[n] = a\,y[n-1] + x[n]$$

where $a = e^{j\theta_0}$ and the frequency of oscillation is θ_0. Let the input $x[n]$ be real and the output $y[n]$ is complex. Thus

$$y[n] = y_R[n] + j\,y_I[n].$$

(i) Determine the equations describing a system with one input $x[n]$ and two outputs $y_R[n]$ and $y_I[n]$ only. [3 Marks]

(ii) Draw a structure for the realisation of this oscillator. [2 Marks]

(iii) Show that if $x[n] = \delta[n]$ then

$$y_R[n] = \cos[n\theta_0].u[n] \text{ and } y_I[n] = \sin[n\theta_0].u[n]$$

[4 Marks]

(iv) Determine the initial conditions required to generate the sinusoids when there is no input. [2 Marks]

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Question 4 (22 marks)

(a)

(i) A second-order analogue band pass filter can be designed from a first-order normalised analogue low pass Butterworth filter by performing the following low pass to band pass transformation:

$$s \rightarrow \frac{s^2 + \omega_0^{\,2}}{B\,s}; \quad B = \omega_u - \omega_l; \quad \omega_0^{\,2} = \omega_u \omega_l$$

where ω_u is the upper cut-off frequency and ω_l is the lower cut-off frequency. Show that the analogue bandpass filter obtained using the above transformation is given by

$$G(s) = \frac{Bs}{s^2 + Bs + \omega_0^{\,2}}$$ [3 marks]

(ii) An IIR digital bandpass filter G(z) is to be designed to approximate the above analogue transfer function G(s). The digital bandpass filter is to operate at a sampling rate of 2 kHz with a lower cut-off frequency (f_l) at 200 Hz and the upper cut-off frequency (f_u) at 300 Hz. Using the bilinear transformation method, obtain the transfer function G(z) of the digital filter and show that G(z) reduces to

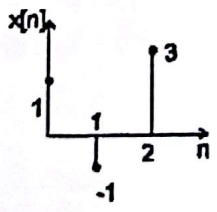
$$G(z) = 0.1367\frac{(1 - z^{-2})}{1 - 1.2362z^{-1} + 0.7265z^{-2}}$$ [5 marks]

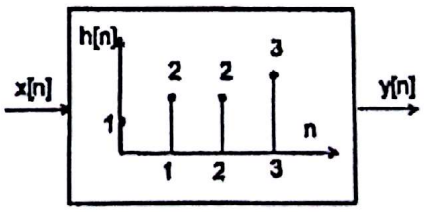
(iii) Draw a canonic realisation of $G(z)$ [3 marks]

(b) Find the transfer function and impulse response of a causal type 3 linear phase filter (assuming the smallest length and smallest delay) if it is known that there is a zero at $z = j$ and two zeros at $z = 1$. (**Note:** A type 3 sequence requires and an odd number of zeros at both $z = 1$ and $z = -1$). [3 marks]

(c) Find the DFT of $x[n] = \{1, 2, 1, 0\}$ given by $X_{DFT}[k]$ where $k = 0,1,2,3$. [4 marks]

(d) The impulse response , h[n], of a discrete time system is given Figure 3. Sketch the output signal, $y[n]$, for the given input signal, $x[n]$.





[4 marks]

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Question 5 (18 marks)

Figure 3(a) shows a modulator scheme where the digital input $x[n]$ is sampled at a sampling period of 8kHz and the magnitude spectrum $|X(\theta)|$ of $x[n]$ is given in Figure 3(b). The carrier frequency of the cosine oscillator $\theta_c = 3\pi/4$.

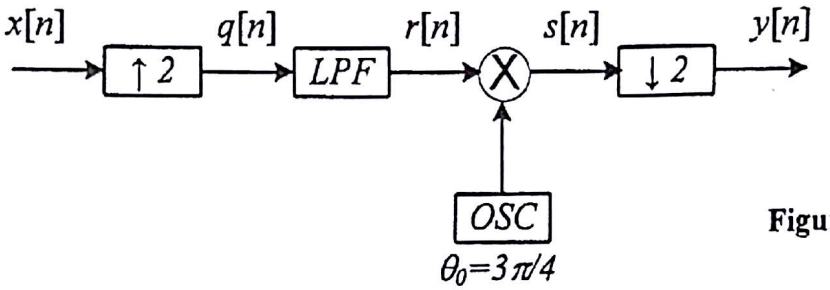


Figure 3(a)

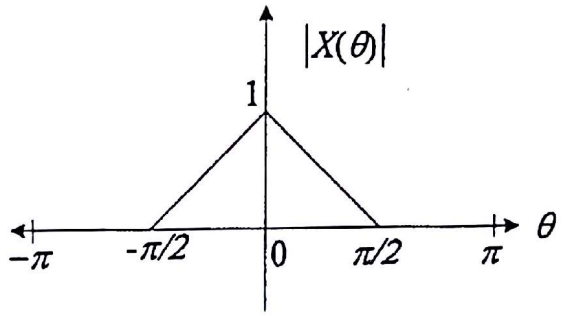


Figure 3(b)

(a) You are required to design a linear phase FIR low pass filter for Figure 3(a). The length of the impulse response is set to $N = 7$ and the coefficients are to be determined using a windowing technique. An ideal low pass filter $H_d(\theta)$, with linear phase can be characterised in the frequency domain by

$$H_d(\theta) = \begin{cases} e^{-j\beta\theta} & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

(i) Determine β. [2 Marks]

(ii) Determine the impulse response $h_d[n]$ [4 Marks]

(iii)Calculate the coefficients of the FIR filter using a rectangular window. [4 Marks]

(b) If $|X(\theta)|$ is the magnitude spectrum of $x[n]$ (as shown in Figure 3 (b), sketch the magnitude spectra of $|Q(\theta)|$, $|R(\theta)|$, $|S(\theta)|$ and $|Y(\theta)|$ in separate diagrams. [8 Marks]

End of examination paper.

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